CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 5: Cyber-Physical Systems & Sensor Networks Integration

Course: Multidisciplinary Minor-III [23CSMDXX]
Category: Assignments | Assignment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Edge computing, embedded sensing protocols, and real-time synchronization in cyber-physical systems.

2. Assignment Problems & Instructions:

1. Evaluate MQTT, CoAP, and HTTP lightweight communication protocols for edge sensors.
2. Design data aggregation algorithms minimizing sensor energy consumption.
3. Analyze clock synchronization PTP IEEE 1588 in distributed telemetry networks.
4. Model closed-loop actuator feedback latency under varying network bandwidths.

Submission Guidelines & Verification:

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